

Fast Food Nutritional Clustering Analysis Project

Project Overview

This project analyzes 722 food items across 37 nutritional attributes (calories, fats, proteins, vitamins, minerals) to discover hidden patterns in food composition. Using unsupervised machine learning, we automatically group similar foods based on their complete nutritional profiles, transforming complex nutritional data into actionable business insights for food manufacturers, retailers, and health-conscious brands.

Business Problem Addressed

Food companies and retailers lack data-driven understanding of how their products cluster nutritionally across the market. This leads to inefficient product development, missed market opportunities, and ineffective marketing to health-conscious consumers. Traditional manual categorization fails to capture complex nutritional relationships that machine learning can uncover.

This project solves this by:

- ✓ Automatically identifying natural nutritional segments across 722 food items
- ✓ Revealing hidden patterns in macro and micronutrient combinations
- ✓ Providing data-driven insights for product portfolio optimization
- ✓ Enabling targeted marketing strategies based on nutritional profiles
- ✓ Identifying market gaps and underserved nutritional segments

Key Performance Indicators (KPIs)

1. **Cluster Quality** – Silhouette Score (target: >0.5 indicating well-separated clusters)
2. **Number of Distinct Segments** – Optimal clusters identified (target: 4-6 actionable segments)
3. **Cluster Interpretability** – Business readability of each cluster (qualitative KPI)
4. **Nutritional Coverage** – Percentage of items well-assigned to clusters (target: >85%)
5. **Actionable Insights** – Number of business recommendations generated (target: 3-5 strategic insights)

Technical Approach

1. **Data Preprocessing**
 - Removed identifier columns (Unnamed indexes)
 - Selected 33 nutritional features for clustering
 - Standardized features using StandardScaler for equal weighting
 - Handled complete dataset with no missing values

2. Exploratory Data Analysis

- Correlation analysis across nutritional dimensions
- Distribution analysis for key nutrients
- Visualized relationships between calories, protein, fat, carbs, and nutrition density

3. Clustering Implementation

- Applied K-Means algorithm for segmentation
- Determined optimal clusters using elbow method and silhouette analysis
- Tested cluster counts from 2-10 to identify natural groupings
- Final selection based on business interpretability and statistical validity

4. Dimensionality Reduction & Visualization

- Used PCA (Principal Component Analysis) to reduce 33 dimensions to 2D and 3D
- Created interactive visualizations for cluster interpretation
- Visualized cluster boundaries and relationships

5. Cluster Profiling & Interpretation

- Calculated mean nutritional profiles for each cluster
- Identified representative foods from each segment
- Mapped clusters to business categories (e.g., "High-Protein," "Nutrient-Dense," "Indulgence")

Conclusion

This project successfully segmented 722 food items into distinct nutritional clusters, revealing patterns that traditional categorization would miss. The analysis identified clear segments such as high-protein foods, nutrient-dense options, high-carb items, and indulgent high-calorie products. These insights enable food companies to optimize product portfolios, develop targeted marketing campaigns, and identify underserved nutritional niches. The clustering framework provides a scalable, data-driven approach to understanding competitive nutritional positioning in an increasingly health-conscious market.

Final Deliverables

- **Clustered Dataset:** Complete dataset with cluster labels (food_nutrition_clusters.csv)
- **Cluster Profiles:** Detailed nutritional profiles for each segment (cluster_profiles_detailed.csv)
- **Visualization Suite:** 8+ visualizations including PCA plots, radar charts, heatmaps, and parallel coordinates
- **Python Scripts:** Reusable code for preprocessing, clustering, and visualization
- **Business Insights Report:** Actionable recommendations for product development, marketing, and portfolio optimization
- **Technical Documentation:** Methodology, feature selection rationale, and cluster interpretation guide